

Analyzing the Impact of Trump's Social Media Sentiment on Crude Oil Price Movements

A business intelligence and machine learning solution that examines whether Donald Trump's social media posts can add useful information when analyzing crude oil price movement, delivering insights through an interactive Power BI dashboard.

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Table of Contents

- Abstract
- Acknowledgment
- Business Intelligence Project Description and Objectives
- Data Research and Acquiring Effort
- Links to Raw Data
- Data Description and Understandings
- Data Primary Cleaning and Transformation
- Data Visualization and Insights
- Exploratory Data Analysis (EDA)
- Dashboard Design and Insights
- Advanced Analytics and AI Modeling
- Sentiment Analysis
- Feature Engineering
- Dataset Merging
- Train/Test Split
- Regression Models
- Classification Models
- Model Evaluation
- Tools Research and Selection Effort

- Project Deployment Effort - Use Case
- Results
- References

Key Results

Metric	Value
Best Classification Model	Sentiment Logistic
Balanced Accuracy	0.5076
Best Regression Model	Dummy Mean
Total Posts Analyzed	32,094
Final Merged Modeling Data	1,200 rows
Sentiment Engine Used	VADER (with TextBlob fallback)

Model Comparison

Model	Type	Performance
Dummy Mean	Regression Baseline	Best R ² among regression models
Sentiment Logistic	Classification	Balanced Accuracy: 0.5076
Oil Baseline Logistic	Classification Baseline	Compared against sentiment models
Sentiment RF	Classification	Compared against baseline

Key Findings

- **Weak Signal Detection:** The project confirms that while Trump's tweets do not control oil prices alone, they can be treated as one possible signal among many other market drivers.
- **Regression vs. Classification:** Regression results were weak, with the Dummy Mean benchmark producing the best R² result. However, classification models performed

slightly better, indicating a weak directional signal.

- **Realistic Framing:** The project is best presented as a realistic weak-signal BI and machine learning study, not as a high-accuracy trading system. It tests association and weak predictive signal, not causality.
- **Data Integration:** Successfully merged 32,094 cleaned tweets with daily crude oil prices, creating a robust dataset of 1,200 daily rows and 228 columns for the period from 2017-02-09 to 2023-03-13.
- **Feature Engineering Value:** The use of lag features, rolling features, and specific keyword detection (oil, energy, geopolitical, trade, sanctions, inflation) was crucial in structuring the noisy public communication into testable indicators.